

Kaustubh Khulbe

408-718-1254 | kkhulbe2@illinois.edu | linkedin.com/in/kaustubh-khulbe | github.com/KaustubhKhulbe | Portfolio

EDUCATION

University of Illinois Urbana-Champaign

August 2023 - May 2026

Bachelor of Science, Computer Science

GPA 3.97/4.00

- **Relevant Coursework:** Advanced OS, Advanced Comp. Arch., Parallel Programming with CUDA, Programming Languages, Algorithms
- **Awards:** Illinois Engineering Achievement Scholar, Dean's List

PROFESSIONAL EXPERIENCE

Incoming Fall 2025 @ NVIDIA

August 2025 - December 2025

Systems Software Engineer Intern - CUDA Performance

San Jose, CA

Amazon Annapurna Labs

May 2025 - August 2025

Compiler Engineer Intern - Graviton Compiler

Austin, TX

- Designing profiling tool to identify hot addresses and lines on a workload.
- Utilizing statistical hardware profiling to create a low-overhead mechanism to monitor load.
- Aiming to integrate software into existing open-source tooling.

University of Illinois at Urbana-Champaign

December 2024 - May 2025

ECE 391 Course Assistant

Urbana, IL

- Developed robust autograder for Machine Projects to allow testing of student code in a sandboxed environment.
- Ran review sessions and office hours for 200+ students.
- Wrote a RISC-V Unix-like OS with paging, filesystem, devices, system calls, and a preemptive scheduling.
- Wrote a `virtio` block driver, cryptography driver, and entropy driver.

Omni RPA Inc.

May 2024 - August 2024

Software Engineer Intern

San Jose, CA

- Built a network transfer cost optimizer for an orchestrator using Go and Nebula.
- Designed large-scale graph models to represent pricing and performance models in Snowflake and Nebula DB.
- Constructed scalable microservices to automate downloading, parsing, and uploading cloud provider data.

RESEARCH EXPERIENCE

SLICE @ Berkeley Architecture Research

May 2025 - Present

RTeAAL — Advised by Chris Fletcher

Berkeley, CA

- Rewriting RTL simulators as tensor problems for new optimization opportunities.
- Formally describing RTL simulation as a cascade of Einsums.
- Writing compiler passes to modify data flow register layouts given FIRRTL Intermediate Representation.
- Creating scheduler to optimize computation graph to reduce heterogeneity layers and reduce branch misdirection.

SLICE @ Berkeley Architecture Research

February 2024 - Present

μSTT — Advised by Chris Fletcher

Berkeley, CA

- Collaboration with Apple to realize STT, a micro-architectural protection mechanism for side-channel attacks.
- Modified RTL and verification to make the RAT (Rename Alias Table) and STT logic stateful.
- Created high-level architecture for a pipelined design to reduce delay.

ADAPT UIUC Lab

November 2024 - Present

ACT (Accelerator Compiler Toolkit) Advised by Charith Mendis

Champaign, IL

- Framework that automates the generation of sound compiler backends for tensor accelerators.
- Evaluated system with synthetic and industry kernels in HLO.
- Wrote a mini transpiler to convert from HLO to Rust and virtual ISA instructions.
- Extensive testing on NKI for Tranium to search for performance related tensor rewrite opportunities.

SKILLS

Programming Languages: C, C++, Rust, Python, CUDA, Triton, HLO, Haskell

Tools and Technologies: Linux/Unix, XLA, QEMU, gdb, nsys/ncu